

CAPSTONE PROJECT CHARTER

Healthcare Operations Intelligence Platform

Predicting Patient Flow Bottlenecks and Optimizing Hospital Resource Allocation Using Operational Analytics

1. Executive Summary

Healthcare organizations operate in highly dynamic environments where patient demand, staffing capacity, and available resources continuously change. One of the most common operational challenges faced by hospitals is inefficient patient flow management, resulting in long waiting times, overcrowded departments, underutilized resources, and increased pressure on healthcare workers.

This project develops a Healthcare Operations Intelligence Platform that applies data analytics, predictive modeling, and interactive dashboarding to improve operational decision-making. The solution analyzes historical patient flow data, department utilization, staffing patterns, and capacity indicators to identify operational bottlenecks and predict future congestion risks.

The final product will be a Streamlit-based analytics application that allows healthcare managers to monitor operational performance, explore trends, identify high-risk areas, and receive data-driven recommendations for resource allocation.

The project demonstrates the application of operational analytics principles including data engineering, exploratory analysis, predictive analytics, dashboard development, and business communication.

2. Industry Domain

Healthcare Operations Management

Healthcare facilities depend on efficient coordination between:

- Patient demand
- Medical staff availability

- Department capacity
- Equipment availability
- Bed utilization
- Appointment scheduling

Although healthcare is the selected application domain, the analytical methods are transferable to other operational environments including energy, manufacturing, logistics, and service industries.

3. Problem Statement

Formal Capstone Structure

In healthcare facilities, the problem of unpredictable patient demand and inefficient resource allocation leads to emergency department congestion, extended patient waiting times, increased staff workload, and reduced operational efficiency.

Currently, hospital managers rely heavily on historical reports and manual monitoring methods, which fail because they provide limited ability to identify emerging operational pressure before service disruption occurs.

We propose using data engineering, predictive analytics, machine learning, and interactive dashboard technology to forecast patient flow challenges, identify operational bottlenecks, and support proactive resource planning.

The project aims to improve operational visibility, reduce waiting times, and enable data-driven decision-making.

4. Business Impact

Poor patient flow creates several operational consequences:

Patient Impact

- Longer waiting times
- Delayed treatment
- Reduced patient satisfaction

Operational Impact

- Uneven staff workload
- Inefficient resource utilization
- Department overcrowding

Financial Impact

- Increased overtime costs
 - Lower operational efficiency
 - Reduced capacity utilization
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5. Project Objectives

The project will achieve the following objectives:

Objective 1

Develop an operational analytics dashboard to monitor hospital performance indicators.

Objective 2

Identify historical patterns affecting patient flow and resource utilization.

Objective 3

Build predictive models to forecast operational pressure.

Objective 4

Provide actionable recommendations to support hospital managers.

Objective 5

Create a deployable Streamlit application demonstrating end-to-end analytics capability.

6. Key Stakeholders

Hospital Operations Managers

Need:

- Capacity visibility
- Department performance monitoring
- Early warning indicators

Department Supervisors

Need:

- Workload information
- Staffing insights
- Bottleneck identification

Healthcare Executives

Need:

- Strategic performance indicators
- Efficiency improvement opportunities
- ROI justification

Data Analysts

Need:

- Reliable operational datasets
 - Automated reporting
 - Analytical tools
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7. Key Performance Indicators (KPIs)

The dashboard will monitor:

Patient Flow KPIs

Average Waiting Time

Measures the average time patients wait before receiving service.

Target:

Reduce average waiting time.

Patient Volume

Measures daily, weekly, and monthly demand patterns.

Length of Stay

Measures operational efficiency after admission.

Department Occupancy Rate

Measures utilization of available capacity.

Formula:

$$\text{Occupancy Rate} = \frac{\text{Occupied Capacity}}{\text{Total Available Capacity}} \times 100$$

Resource KPIs

Staff Utilization Rate

Measures whether staffing levels match operational demand.

Capacity Risk Score

Predicts whether a department is approaching operational overload.

8. Data Sources

The project will use a combination of public and realistically generated operational data.

Source 1: Public Healthcare Datasets

Potential sources:

- MIMIC-IV clinical and hospital dataset
- Public emergency department utilization datasets

Purpose:

Provide realistic healthcare operational structures.

Source 2: Synthetic Hospital Operations Dataset

A realistic operational layer will be developed.

Example variables:

Field	Description
Patient_ID	Unique patient identifier
Date	Visit date
Department	Service area
Arrival_Time	Patient arrival time
Treatment_Time	Service duration
Staff_Count	Available staff
Bed_Count	Department capacity
Occupied_Beds	Current utilization
Waiting_Time	Delay duration
Admission_Status	Outcome

9. Technology Stack

Programming

Python

Libraries:

- Pandas
 - NumPy
 - Scikit-learn
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Database

SQLite / PostgreSQL

Purpose:

Store processed operational datasets.

Visualization

- Plotly
 - Matplotlib
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Dashboard

Streamlit

Features:

- Interactive filters
 - KPI cards
 - Dynamic charts
 - Prediction interface
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Machine Learning

Models:

Forecasting

- Random Forest Regression

Classification

- Logistic Regression
 - Random Forest Classifier
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10. Analytics Workflow

Phase 1: Data Collection

Collect:

- Patient activity data
- Department information
- Resource availability

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Phase 2: Data Cleaning

Tasks:

- Missing value treatment
- Data validation
- Duplicate removal

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Phase 3: Exploratory Data Analysis

Analyze:

- Patient trends
- Waiting time patterns

- Department performance



Phase 4: Predictive Modeling

Develop:

- Demand forecasting model
- Congestion risk model



Phase 5: Dashboard Development

Create:

- Executive dashboard
- Operational dashboard
- Prediction interface



Phase 6: Business Recommendations

Translate findings into:

- Staffing recommendations
- Scheduling improvements
- Capacity planning actions

11. Success Metrics

The project success will be measured by:

Technical Metrics

Machine learning:

- Model accuracy
- Precision

- Recall
 - F1-score
 - Prediction error
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Operational Metrics

Target improvements:

- Improved visibility of patient demand
 - Earlier identification of congestion risk
 - Better resource planning decisions
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Dashboard Metrics

Successful completion:

- Interactive filtering
 - Automated KPI calculations
 - Real-time risk visualization
 - Prediction capability
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12. Project Timeline

Week 6: Project Foundation

Deliverables:

- Finalize problem statement
- Define KPIs
- Identify datasets
- Create project architecture

Week 7: Dashboard Development

Deliverables:

- Build Streamlit framework
- Create operational dashboards
- Visualize healthcare KPIs

Week 8: Operational Storytelling

Deliverables:

- Analyze bottlenecks
- Develop stakeholder insights
- Create executive narrative

Week 9: Machine Learning Integration

Deliverables:

- Train predictive models
- Evaluate performance
- Analyze feature importance

Week 10: AI-Assisted Product Development

Deliverables:

- Add AI assistant capability
- Improve dashboard interaction
- Automate reporting

Week 11: Strategy and Presentation

Deliverables:

- ROI analysis
 - Final report
 - Stakeholder presentation
 - Portfolio preparation
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